What the VA Prazosin Trial CSP 563 Can Tell the pmsimstats-ng Simulations: Empirical Checks and Candidate Calibrations

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Author. pmsimstats team

Purpose. The pmsimstats-ng simulations rest on a data-generating process (DGP) whose parameters were set from the published code of Hendrickson et al. (2020), which in turn calibrated them to an earlier positive trial of prazosin in active-duty soldiers. This paper asks which of those parameters can be checked or estimated against participant-level data from a second, larger trial: the VA Cooperative Studies Program trial CSP 563 (Raskind et al. 2018, 304 veterans, negative overall). It reports what was estimated, what could not be, and what each result implies for the simulations and the manuscripts.

Source. All analyses were run in the private compendium `~/prj/alz/14-murray-nejm/murraynejm`, branch `sbp-caps-replication`, directory `analysis/report/04-sbp-caps-replication/`. Two analysis plans were committed before any model in them was fitted (`plan.md`, `plan-components.md`); a measurement-error follow-up and the parameter summaries were added afterwards and are labeled as such. Only aggregate results appear here. The participant-level data stay in that compendium. The studies are exploratory.

Evidence labels. *Computed* means produced by a script in this session from the CSP 563 data or from the pmsimstats-ng formulas. *Quoted* means taken from a publication. *Inferred* means argued from the computed results and not itself tested.

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Notation

Symbols follow `analysis/report/NOTATION.md`.

Table 1: Notation

Symbol	Meaning
Y_{it}	CAPS-IV total symptom severity of participant i at visit t
BL_i	baseline severity; the DGP writes $Y_{it} = BL_i - (TV_{it} + PB_{it} + BR_{it})$
TV, PB, BR	natural-history, placebo-response and drug-response components
η_t	design expectancy: 1 open label, 0.5 blinded
B_i	biomarker: standing systolic blood pressure (SBP), mm Hg
σ_{BR}, σ_{bm}	DGP standard deviations of the drug response (8) and the biomarker
c_{bm}	correlation between the biomarker and the drug response (the interaction strength)
$c_{bm,BL}$	correlation between the biomarker and baseline severity
ρ, c_1	DGP AR(1) correlation per week within a component (0.7) and between components (0.2)
φ, ν	fitted serial correlation per week and measurement-error share of the CAPS total
τ	SD of individual drug effects (treatment heterogeneity)
$a^\top Aa$	time part of the variance of a participant's on-minus-off contrast

Glossary

- **CSP 563.** VA Cooperative Studies Program trial of prazosin for PTSD in veterans; 304 randomized, 10-week primary endpoint, 26-week follow-up; no benefit on its primary outcomes (Raskind et al. 2018).
- **CAPS-IV total.** Clinician-Administered PTSD Scale for DSM-IV, total symptom severity; the outcome scale of the DGP. A secondary outcome of CSP 563.
- **Correlation encoding.** The DGP's encoding of the biomarker-by-treatment interaction as a correlation between the biomarker and the drug-response component.
- **CR2.** Bias-reduced cluster-robust standard error with Satterthwaite degrees of freedom.
- **Extended DGP.** The pmsimstats-ng recommended DGP, with separable AR(1) correlation ($\rho = 0.7$, $c_1 = 0.2$).
- **Nugget.** Visit-specific measurement error: variance not shared with any other visit.
- **Predictive, prognostic.** A biomarker that modifies the drug effect, and one associated with the outcome regardless of drug.

1. Summary

1. **The blood-pressure biomarker effect of Raskind et al. (2016) does not replicate.** In 279 participants the arm-by-SBP interaction on CAPS total change was +0.25 points per 10 mm Hg (95% CI −2.72 to 3.21; computed). The prazosin-arm slope excludes the published −14.
2. **CSP 563 cannot test the effect sizes the simulations use.** On the simulation scale the interval is c_{bm} −0.57 to 0.48. The project's grid, c_{bm} 0.25 to 0.45, implies only 1.4 to 2.5 points per 10 mm Hg, inside the interval. The published −14 would need $c_{bm} \approx 2.5$, which is impossible for a correlation (computed).
3. **The biomarker is not prognostic.** SBP and baseline CAPS correlate at $r = 0.02$ (−0.09 to 0.13; computed). The DGP's $c_{bm,BL} = 0$ is supported.
4. **CAPS correlation decays slowly, with about 10 percent measurement error.** Compound symmetry fits worst. The total-score correlation implied by the extended DGP falls too fast at short lags and then flattens (computed).
5. **Under that covariance the Hybrid design's advantage over the crossover is larger** than under the extended DGP: contrast noise 0.37 to 0.41 of the crossover's, against 0.59 to 0.69 (computed).

6. **Placebo-arm improvement is about 1.8 times the DGP's, with the same time course.** There is no mean drug response, and the SD of individual drug effects is at most about 8 points (computed).
7. **Missingness was unrelated to prior worsening**, unlike the DGP's biased dropout (computed).
8. **Natural history and placebo response cannot be separated** in a fully blinded parallel trial. Neither can the carryover half-life or the per-component correlations.

2. The data and what the design identifies

CSP 563 randomized 304 veterans with chronic PTSD to prazosin or placebo, blinded throughout (Raskind et al. 2018, quoted). CAPS total was recorded at screening and weeks 6, 10, 18 and 26 (303, 273, 263, 249 and 245 records). Standing and supine blood pressure were recorded at screening, on randomization day and at most later visits.

The design fixes what can be learned (inferred from the design):

- Everyone was blinded at every visit, so $\eta_t = 0.5$ throughout and time under expectancy equals calendar time. TV and PB share one clock and one weight and are **not separately identifiable**.
- No one stopped drug during the trial, so there is **no carryover information**.
- It is a parallel trial: the drug response is identified only as the difference between arms, and its variability only through the difference in outcome variance between arms.
- The data export has no expectancy or treatment-guess measure and no study-drug dose records.

3. The biomarker

3.1 Predictive effect

Mixed model of CAPS total change at weeks 6 to 26 with arm, SBP and week and their interactions, adjusted for baseline CAPS, the two randomization stratifiers and site; unstructured covariance; CR2 inference. SBP is the mean of the screening and randomization-day readings (computed).

Table 2: SBP-by-arm interaction on CAPS total, 279 participants

Quantity, CAPS points per 10 mm Hg	Estimate	95% CI
Interaction (prazosin minus placebo slope)	+0.25	−2.72 to 3.21
Prazosin-arm slope	−0.89	−3.14 to 1.35
Placebo-arm slope	−1.14	−3.23 to 0.95
Interaction, week 10	+0.09	−4.85 to 5.03

The result did not depend on the analysis choices (computed). The interaction was −0.51 (−3.06 to 2.04) with screening SBP only, +1.22 (−1.41 to 3.86) with randomization-day SBP only, −0.25 (−3.80 to 3.29) with supine SBP, and +0.35 (−3.00 to 3.70) in a week-10 ANCOVA.

3.2 Translation to the simulation scale

Under the correlation encoding the arm-by-biomarker slope is $-c_{bm}\,\sigma_{BR}/\sigma_{bm}$ per biomarker unit (`docs/39`). With $\sigma_{BR} = 8$ and the CSP 563 biomarker SD of 14.2 mm Hg (computed):

Table 3: The interaction on the c_{bm} scale

Quantity	Value
CSP 563 interval for c_{bm}	−0.57 to 0.48
Slope implied by $c_{bm} = 0.25$	−1.41 per 10 mm Hg
Slope implied by $c_{bm} = 0.45$	−2.54 per 10 mm Hg
c_{bm} implied by Raskind 2016 (−14 per 10 mm Hg)	about 2.5

The project's effect sizes are an order of magnitude below the published one and are not excluded by CSP 563. The published effect cannot be represented by the DGP at all, because a correlation cannot exceed 1 (inferred from the formula). With a smaller drug-response SD the interval in c_{bm} widens: at $\sigma_{BR} = 5$ it is −0.91 to 0.77.

Implication. Manuscripts that cite the 14-points-per-10-mm-Hg association as motivation should state that it was not replicated and that the simulated effects are far smaller. The simulated grid stays defensible as a range of hypothetical biomarker effects. No trial has yet estimated c_{bm} for SBP with useful precision.

3.3 Prognostic association

Table 4: Biomarker and baseline severity (computed)

Comparison	n	Pearson r (95% CI)
Standing SBP vs screening CAPS	303	0.02 (−0.09 to 0.13)
Supine SBP vs screening CAPS	303	0.01 (−0.10 to 0.12)
Placebo arm: standing SBP vs week-10 CAPS change	132	−0.07 (−0.24 to 0.10)

Correlations larger than about 0.13 in absolute value are excluded. The DGP's $c_{bm,BL} = 0$ is consistent with these data, and the `prog-BL` scenarios of paper 02 and `docs/64` remain sensitivity analyses rather than expected conditions.

4. Covariance of the CAPS total

4.1 Estimates

Placebo arm, 152 participants, CAPS total at weeks 0, 6, 10, 18 and 26 (computed):

Table 5: Empirical correlation of CAPS total, placebo arm

	0	6	10	18	26
0	1	0.63	0.60	0.55	0.46
6		1	0.82	0.77	0.64
10			1	0.84	0.73
18				1	0.81
26					1

Table 6: Covariance models for CAPS total, placebo arm (computed; the nugget model was added after the plan)

Model (REML, week-factor mean)	ΔAIC	Parameters
Unstructured	0	as the table above
Exponential decay + nugget	39	$\nu = 0.10$, $\varphi = 0.975$ per week
Random intercept + exponential	50	intercept share 0.40, $\varphi = 0.924$
Exponential (AR(1))	52	$\varphi = 0.959$
Compound symmetry	110	intraclass correlation 0.68

The nugget model implies correlations of 0.88, 0.82, 0.74 and 0.55 at lags of 1, 4, 8 and 20 weeks, against observed 0.82 to 0.84 at 4 weeks, 0.77 at 8 and 0.64 at 20. Screening correlates less with every later visit than follow-up visits do with each other. Selection on screening CAPS for eligibility and regression to the mean are plausible explanations; this was not tested.

4.2 Comparison with the extended DGP

The extended DGP's total-score correlation, with BL SD 18.48, component SDs 10, $10\eta_t$ and 8, $\rho = 0.7$ and $c_1 = 0.2$, is 0.87 at 1 week, 0.67 at 4 weeks and about 0.57 from 8 weeks on (computed). It decays faster than CAPS at short lags and then flattens, where CAPS keeps decaying. Matching the observed 4-week and 20-week correlations within the DGP's structure would require ρ of about 0.87 to 0.91 per week (computed).

The variance also differs. The DGP's SD of change from baseline is 16.0 at every visit. In the CSP 563 placebo arm it is 17.0, 19.4, 22.1 and 24.4 at weeks 6, 10, 18 and 26 (computed).

4.3 Consequence for design comparisons

The time part of the on-minus-off contrast variance, $a^\top A a$ with $A = (1-\nu)\varphi^{|w_t - w_s|} + \nu I$ (computed):

Table 7: Contrast noise by design and covariance structure

Structure	Crossover	Hybrid paths	Alternating	Hybrid / crossover
Extended DGP, $\rho = 0.7$, no noise	0.770	0.451 to 0.531	0.235	0.59 to 0.69
`docs/39`, $\rho = 0.8$, 20% noise	0.763	0.389 to 0.466	0.232	0.51 to 0.61
CSP 563, $\varphi = 0.975$, 10% noise	0.311	0.115 to 0.128	0.074	0.37 to 0.41

Under the CSP 563 structure every within-patient contrast is less noisy, and interleaving on and off visits helps more. The design rankings of `docs/39` and `docs/68` Section 9 under AR(1) are, if anything, understated (inferred). Two limits apply. CSP 563 assessed CAPS every 4 to 8 weeks, so correlations at the Hybrid design's one-week lags are extrapolated from the fitted form. And the calculation covers the noise of the contrast only, not carryover or the proportional-moderation question of `analysis/scripts/quick-sim/carryover-closed-form/22-proportional-moderation-designs.R`.

5. Response components

5.1 Mean trajectories

Adjusted mean CAPS change, CR2 95% CIs, against the DGP's means under blinding (computed):

Table 8: Mean CAPS change (negative is improvement)

Week	Placebo ($TV + PB$)	DGP	Prazosin minus placebo (BR)	DGP
6	−9.7 (−13.1 to −6.3)	−5.3	−1.1 (−5.0 to 2.8)	−7.3
10	−13.0 (−16.8 to −9.2)	−8.4	+1.1 (−3.3 to 5.5)	−10.2
18	−17.7 (−21.9 to −13.6)	−9.7	+4.4 (−0.6 to 9.3)	−11.0
26	−17.5 (−22.1 to −12.9)	−9.8	+1.5 (−4.1 to 7.2)	−11.0

A modified Gompertz curve fitted to the placebo means, with the displacement fixed at 5, gives a maximum of 17.2 points (SE 0.9) and a rate of 0.34 per week (SE 0.03). The DGP's $TV + 0.5\,PB$ has maximum 9.8 and rate 0.35 (computed). The shape matches and the size is about 1.8 times larger. How the larger improvement divides between TV and PB is not identifiable here.

5.2 Drug-effect heterogeneity

Residual SD of CAPS change by arm (week-by-week ANCOVA, 2,000 bootstrap resamples within arm; computed):

Table 9: Outcome SD by arm and the implied bound on τ

Week	SD placebo	SD prazosin	Ratio (95% CI)	$\hat{\tau}$ upper 95% limit
6	16.1	16.1	1.00 (0.81 to 1.24)	10.2
10	19.0	17.5	0.92 (0.75 to 1.12)	8.3
18	20.7	18.2	0.88 (0.71 to 1.08)	7.0
26	23.3	21.7	0.93 (0.76 to 1.14)	11.0

A longitudinal test of arm-specific variances gave $\chi^2 = 3.87$, 4 df, p = 0.42. The bound assumes individual drug effects are uncorrelated with the other components.

5.3 Blood-pressure fall as a marker of drug action

Prazosin lowered supine SBP by 5.7 mm Hg relative to placebo at week 10 (−8.8 to −2.6). Within the placebo arm a larger fall in SBP went with greater CAPS improvement (3.1 points per 10 mm Hg, 0.4 to 5.8). The relation was not stronger in the prazosin arm (interaction −2.0, −5.7 to 1.7; computed). A fall in blood pressure tracks a shared state, not individual drug response. SBP change is post-randomization; the analysis is descriptive.

6. Dropout

The share of participants seen at one visit but not the next was 10 to 11 percent from screening to week 6 and 3 to 8 percent in later intervals, similar between arms. In a logistic model over the post-baseline intervals, CAPS worsening so far did not predict missing the next visit: odds ratio 0.96 per 10 points of worsening (0.82 to 1.13; computed). The DGP's `censordata()` makes dropout depend on worsening (corrected 2026-10-29). CSP 563 shows no such dependence. The count includes intermittent missed visits as well as permanent dropout.

7. Parameter table

Table 10: DGP parameters against CSP 563

DGP parameter	Current value	CSP 563	Status
BL mean, SD	83.1, 18.5	81.1, 16.0	estimated
Biomarker mean, SD	124.3, 15.4	129.8, 14.2	estimated
$c_{bm,BL}$	0	0.02 (-0.09 to 0.13)	estimated; supports current value
c_{bm}	grid 0.25 to 0.45	-0.57 to 0.48	not informative at the grid's scale
BR mean	Gompertz maximum 11.0	about 0	estimated for this trial
$TV + PB$ under blinding	maximum 9.8, rate 0.35	maximum 17.2, rate 0.34	estimated in combination
TV versus PB split	equal (assumption)	none	not identifiable
Expectancy weight 0.5	assumption	none	not identifiable
Component SDs (10, 10η, 8)	fixed	SD of change 17 to 24, growing	total only
ρ, c_1	0.7, 0.2	total score: $\varphi = 0.975$, $\nu = 0.10$	total only
τ (paper 04)	grid	at most about 8 at week 10	bounded
Dropout dependence on worsening	yes	none detected	estimated
Carryover half-life	0.1 to 0.2 week	none	not identifiable
Proportional moderation	not used	none	not identifiable

8. Implications for the manuscripts

- **Papers 01 and 02 (DGP and carryover).** The cited 14 points per 10 mm Hg did not replicate; say so, and note that the simulated c_{bm} grid is an order of magnitude smaller and is not excluded. $c_{bm,BL} = 0$ now has empirical support.
- **Paper 04 (main effect).** The drug-response mean of 11 points describes the earlier positive trial; CSP 563 found none. Check the heterogeneity grid against the bound of about 8 points.
- **Paper 05 and `docs/68` (design).** The CSP 563 covariance favors interleaving more than the extended DGP does. A sensitivity arm at $\varphi \approx 0.97$ per week with 10 percent noise would test whether the design rankings hold.
- **Paper 09 (dropout).** CSP 563 gives no support for worsening-dependent dropout. A missing-completely-at-random arm is the empirically closer condition for veterans in a blinded trial.
- **Paper 10 (calibration).** Paper 10 traced κ above 1 mainly to mean-model misfit. The growing variance of change seen here is a second, untested candidate (inferred).

9. A candidate CSP 563-calibrated sensitivity arm

If the project adds one sensitivity DGP, these values follow from the estimates (inferred; none has been simulated):

1. BL SD 16; biomarker SD 14.
2. TV and PB maxima each scaled by 1.8, keeping the equal split (an assumption this trial cannot test) and the rate 0.35.
3. Component correlation decaying at about 0.97 per week, plus a measurement-error component of about 10 percent of total variance. The per-component split remains an assumption.
4. Component variances growing over time so that the SD of change rises from about 17 to 24 over 26 weeks.
5. Dropout independent of worsening, about 3 to 11 percent per interval, highest in the first six weeks.
6. c_{bm} grid unchanged; $c_{bm,BL} = 0$.
7. BR mean at 0 for a CSP 563-like scenario and at 11 for the earlier trial, reported side by side.

10. Limitations

- One trial, negative overall, in chronic veterans; the DGP's calibration source enrolled active-duty soldiers. The two may differ in placebo response and drug sensitivity.
- CAPS total was a secondary outcome of CSP 563.
- Visits 4 to 8 weeks apart limit what can be said about one-week lags.
- The heterogeneity bound and the component means assume additivity and independence of the drug response from the other components.
- The plans were committed before fitting but not registered; the nugget models, the parameter summaries and the dropout analysis were added afterwards.

References

- Hendrickson RC, Thomas RG, Schork NJ, Raskind MA (2020). Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2:13. doi:10.3389/fdgth.2020.00013.
- Raskind MA, Millard SP, Petrie EC, et al. (2016). Higher pretreatment blood pressure is associated with greater posttraumatic stress disorder symptom reduction in soldiers treated with prazosin. *Biological Psychiatry* 80(10):736-742. doi:10.1016/j.biopsych.2016.03.2108.
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- Project documents: `docs/39` (design optimization), `docs/40` (Hybrid design primer), `docs/64` and paper 02 (prognostic scenarios), `docs/68` (design tutorial), papers 04, 09 and 10.
- Analysis sources: `~/prj/alz/14-murray-nejm/murraynejm/analysis/report/04-sbp-caps-replication/` (`plan.md`, `results.md`, `plan-components.md`, `results-components.md`, scripts `01` to `04`).